Gain–Loss Asymmetry and Bimodal Structures in Multiscale Financial Returns: Disentangling Market Dynamics from Random Walks

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Abstract

This study presents a multi-scale statistical analysis of financial market dynamics by examining "Trend Returns" (TReturns)—logarithmic returns derived from uninterrupted directional price movements—across four major global indices (DJIA, Nasdaq, Nikkei, and IPC). By contrasting empirical data against a Monte Carlo Geometric Brownian Motion (GBM) null model, we rigorously distinguish between mechanical artifacts and genuine structural anomalies. We document Emergent Bimodality: while daily returns ($d = 1$) exhibit unimodal noise, trends persisting for $d \geq 2$ structurally separate into distinct positive and negative modes. While the GBM confirms that this separation is largely a natural topology of trend persistence, empirical data reveal a distinct divergence at longer durations ($d > 6$), where the negative mode extinguishes, transitioning back to a positively shifted unimodality—a behavior absent in the persistent bimodality of the random walk. Furthermore, we uncovered a fundamental Dual Asymmetry. While the Asymmetry of Frequency (preference for long uptrends) is partially consistent with positive drift, the Asymmetry of Character—quantified via Volatility Ratios (< 1) and the Wasserstein Distance—reveals that downtrends are structurally more volatile ("violent") and possess heavier tails than uptrends, a feature that the Gaussian control fundamentally fails to reproduce. At the macro level, this manifests as a universal negative skewness. These findings suggest that standard Gaussian models are structurally blind to the specific mechanics of market corrections, leading to systematic underestimation of tail risk.

Keywords: Market Trends, Emergent Bimodality, Gain-Loss Asymmetry, Wasserstein Distance, Multi-scale Analysis, Financial Stylized Facts

1. Introduction

Understanding the statistical behavior of financial returns is the cornerstone of quantitative finance, serving as the foundation for asset pricing[1, 2, 3], risk management [4], and algorithmic trading strategies [5]. The classical framework, built upon the Efficient Market Hypothesis (EMH) [6] and the Random Walk model [7], conventionally assumes that price changes follow a Gaussian distribution[2, 8] and are independent over time.

However, decades of empirical research have established a set of "stylized facts"—such as heavy tails (leptokurtosis), volatility clustering, and gain-loss asymmetry—that the normal distribution fundamentally fails to capture [9, 10]. To address these discrepancies, the literature has largely focused on probabilistic models that incorporate deviations to capture and emulate particular stylized facts in the market [11, 12, 13, 14]. Although powerful, these approaches typically analyze returns over fixed chronological intervals (e.g., daily or monthly closing prices), sharing the premise that returns fluctuate around a single equilibrium point—typically zero. From this perspective, price changes are essentially noise around a central tendency. We argue that while fixed-time analysis is still fundamental, it can mask the market's inherent "intrinsic time" and make it challenging to clearly separate stochastic fluctuations from authentic directional trends.

In this study, we adopt a multiscale approach based on "Trend Returns" (*TReturns*). These returns emerge from the concept of "price runs"—time intervals during which the price of a financial asset moves upward or downward without interruption, indicating a clear market trend. These returns are computed across multiple temporal scales, allowing for the analysis of trend dynamics across various time horizons. Unlike traditional returns, multiscale returns enable the identification and quantification of persistent directional movements, offering a more flexible lens through which to study price behavior in the stock market. The relevance of this approach lies in its ability to capture the temporal structure of financial time series, which often display trends and reversal patterns influenced by investor behavior [15] or market dynamics [16]. Additionally, by accounting for multiple timescales, this method enhances the detection of nonlinear patterns and adaptability to markets with varying degrees of volatility.

Using historical data from four major indices (DJIA, Nasdaq, Nikkei, and IPC) alongside synthetic data generated from a Monte Carlo GBM simulation, which serves as a null validation model, we report three primary contributions that expand the understanding of financial time series.

First, we systematically characterized the emergence and evolution of bimodality. We demonstrate that, after conditioning *TReturns* on the trend duration, the resulting distribution exhibits a structural pattern consistent with theoretical expectations. Specifically, whereas short-term fluctuations ($d = 1$) yield a unimodal distribution, trends persisting for $d \geq 2$ display a statistically significant bimodal distribution. This bimodal separation reflects the intrinsic topology of persistent trends and is accurately replicated by the geometric Brownian motion (GBM) null model. However, the empirical data exhibit a distinct divergence at longer durations ($d > 6$), where the distributions transition back toward a shifted unimodal form. This effective "extinction" of the negative mode in real markets contrasts sharply with the persistent bimodality observed under the random-walk benchmark. Moreover, the statistical properties of these emergent modes indicate that the "bull" and "bear" regimes—typically modeled as latent states—admit a directly observable and quantifiable structural counterpart once high-frequency (daily) noise is filtered out.

Second, we uncover a fundamental, dual asymmetry. On the one hand, an Asymmetry of Frequency: in most markets, long-horizon trends are disproportionately likely to be positive (a phenomenon linked to negative mode extinction), a feature that our GBM benchmark also reproduces and can be partly attributed to its positive drift. On the other hand, markets display a second, more consequential irregularity: an Asymmetry of Character. Using the Wasserstein Distance and moment-based ratios, we demonstrate that negative trends are intrinsically more volatile—i.e., more "violent"—than positive trends of comparable scale.This microscopic-level asymmetry produces a systematically negative skewness in the aggregated distribution, a feature that a symmetric Gaussian specification is, by construction, incapable of reproducing.

Finally, we demonstrate the universality of these properties across diverse regulatory environments. By contrasting these stylized facts against the GBM benchmark, we isolate the structural properties of markets that standard Gaussian models fail to capture: specifically, the extinction of the negative mode for larger trends (d>6), the asymmetry of character (volatility), the negative skewness persistent across trend durations, and a decrease in kurtosis as a function of trend duration. Thus, we establish rigorous empirical benchmarks for validating future synthetic market models.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature. Section 3 details the data and methodologies. Section 4 presents the empirical results on bimodality and asymmetry. Finally, Section 5 discusses the theoretical and practical implications of our findings.

2. Literature Review

A systematic review of the quantitative finance literature reveals a coherent set of stylized facts that challenge the premises of traditional asset pricing models. This section synthesizes the main academic currents relevant to our multi-scale analysis, focusing on four key axes: the multi-scale approach, the inadequacy of the normal distribution, gain-loss asymmetry and trend persistence.

2.1. The Multi-Scale Approach in Finance

Analyzing financial time series across different time horizons is a methodological necessity driven by the complex statistical properties of markets [9]. Within this domain, distinct approaches have been developed. One prominent stream is "inverse statistics," which fixes a return target and analyzes the distribution of waiting times required to cross it [17, 18].

Complementary to this, yet methodologically distinct, is the approach followed in this study: analyzing the returns of directional price trends ("runs") across varying durations. Building upon the methodological framework established in previous works within this research line, specifically, Trend Returns ("TReturns") and duration-normalized Trend Returns ("TVReturns") derived from daily uninterrupted trends [19, 20], we extend the analysis of multi-scale financial variables constructed from uninterrupted price movements.

These observables are inherently multi-scale and advantageous for studying market symmetry, as they contain a nearly identical number of positive and negative terms [19]. Furthermore, empirical analysis of price trend durations in major indices, such as the DJIA, Nasdaq, and IPC, has been used to evaluate the geometric distributional behavior of the market, offering a new perspective on the Efficient Market Hypothesis [20]. Recent formal studies have concluded that despite long-memory oscillations in the Autocorrelation Function (ACF), these multi-scale return series pass stationarity tests (e.g., Augmented Dickey-Fuller), allowing them to be treated as mean-stationary processes despite being conditionally heteroscedastic [21].

2.2. The Inadequacy of the Normal Distribution

The economic and econophysics literature has established that financial asset returns are characterized by a set of stylized facts that the normal (Gaussian) distribution cannot capture. This presents a challenge to the classical Random Walk model, tracing its origins to Bachelier's seminal theory of speculation [22], which posited an Arithmetic Brownian Motion for absolute price changes. This framework was later refined by Samuelson [23], who introduced the Geometric Brownian Motion (GBM) to model percentage returns and ensure price non-negativity—a paradigm subsequently formalized by Fama [6] under the Efficient Market Hypothesis. However, pioneering work by Mandelbrot [10] identified that speculative prices exhibit "heavy tails," a phenomenon that fundamentally violates these Gaussian assumptions.

A fundamental characteristic is leptokurtosis, which implies a significantly higher probability of observing large price changes than that predicted by normal models [9, 18]. The kurtosis of financial indices typically exceeds 3, although it tends to decay as the sampling interval increases [24]. In addition, the conditional return distributions consistently exhibit skewness and asymmetry. While early studies linked this to volatility feedback [25], recent research continues to document persistent asymmetry in the distributions of accumulated gains and losses [26], with skewness coefficients often significantly different from zero [27].

To model these features, the literature first evolved towards time-varying volatility frameworks, initiated by Engle's Autoregressive Conditional Heteroscedasticity (ARCH) model [28] and subsequently expanded by Bollerslev into Generalized ARCH (GARCH) models [11]. While these approaches address volatility clustering, they often require non-normal conditional distributions, such as the standardized or skewed Student's t distribution, to match the empirical tails [27, 29]. Complementary to this, to specifically address non-normality and trend persistence, the literature has relied on regime-switching frameworks, particularly Hidden Markov Models (HMM). These probabilistic generative models aim to infer latent state sequences (market regimes) from observed returns, with evidence often favoring the switching behavior at the index level [13, 30].

Further refinements have led to continuous-time stochastic models that incorporate stochastic volatility and discontinuous jumps. Recent comparative analyses, such as Kahssay and Miah [14], demonstrate that while advanced models such as Stochastic Volatility with Jumps (SVJ) provide superior forecasting performance by capturing these nonlinear dynamics, the Geometric Brownian Motion (GBM) remains the essential baseline against which these improvements are measured. This reinforces the utility of the GBM not as a perfect descriptor of reality but as the canonical "null model" for isolating structural anomalies.

Finally, the field is currently expanding towards hybrid and generative approaches. Studies such as Kim and Won [12] have successfully integrated Long Short-Term Memory (LSTM) networks with GARCH-type models to enhance volatility forecasting, while Takahashi et al. [31] have employed Generative Adversarial Networks (GANs) to reproduce complex stylized facts. However, even these advanced data-driven methods require rigorous statistical benchmarks to ensure that they capture the fundamental asymmetries of the market rather than merely overfitting noise.

2.3. Gain-Loss Asymmetry

Gain-loss asymmetry is a robust stylized fact manifested in both volatility dynamics and investor behavior. A primary manifestation of this is the Leverage Effect, a phenomenon originally described by Black [32], which involves a negative correlation between an asset's returns and its future volatility. Specifically, empirical evidence shows that volatility tends to increase significantly following price drops or negative returns [33, 27]. While classical explanations attribute this to increased financial leverage as a firm's equity value falls [34], this mechanism is often insufficient to explain the magnitude of the effect at the aggregate market level [35]. This has led to alternative hypotheses involving "volatility feedback" and market panic [33]. This systemic view was further developed by Sornette [16], who framed extreme negative regimes not merely as exogenous shocks but as critical phase transitions driven by endogenous positive feedback loops. This provides a physical mechanism for explaining why market corrections exhibit structural "violence" distinct from the accumulation phase.

Complementing this mechanical and systemic view, Behavioral Finance offers a psychological explanation rooted in Prospect Theory [15]. This framework posits that economic agents exhibit loss aversion, being significantly more sensitive to losses than to gains of equal magnitude, and often display risk-seeking behavior in the loss domain [36]. Models incorporating these behavioral elements have demonstrated greater explanatory power for market anomalies [37]. Crucially, the leverage effect, endogenous instability, and loss aversion all converge to predict a similar structural asymmetry in return distributions: "heavier" tails and higher volatility for downward movements than for upward trends.

2.4. Persistence and Momentum

Beyond asymmetry in magnitude, financial time series exhibit significant anomalies in terms of temporal persistence. Return persistence, widely recognized as momentum, challenges the weak form of the Efficient Market Hypothesis [38]. Specifically, there is strong evidence of Time Series Momentum (TSMOM), where an asset's past returns positively predict its future trend direction [39].

This effect is consistent across diverse asset classes and typically persists for horizons between one and twelve months. While some studies propose macroeconomic risk explanations [40], global evidence suggests that momentum profitability remains statistically significant across varying economic states [41], lending support to behavioral models focused on imperfect investor expectation formation and under-reaction to information [42]. Notably, TSMOM exhibits a "straddle-like" behavior, generating its largest profits during extreme market episodes [39], suggesting that trend persistence is a fundamental feature of market mechanics rather than a transient anomaly.

3. Data and Methodology

3.1. Data

This study analyzes the daily closing prices of four major global stock market indices sourced from *Yahoo Finance API* to assess the universality of trend-return patterns.

- Dow Jones Industrial Average **(DJI):** Representing 30 large blue-chip companies listed in the United States. (Period: 1992-2025).
- Nasdaq Composite **(Nasdaq):** Focused on the technology and growth sector of the U.S. market. (Period: 1971-2025).
- Nikkei 225 **(Nikkei):** The primary index for the Japanese market, sensitive to exports and technology. (Period: 1970-2025).
- Indice de Precios y Cotizaciones **(IPC):** The main indicator of the Mexican market, representing the largest and most liquid firms of the country; important case of study as an emergent market. (Period: 1991-2025).

The data span several decades and encompass multiple market cycles, crises, and calm periods. A descriptive summary of the time series and identified trends is presented in Table 1. We utilize the full available historical history for each index rather than synchronizing the start date. This approach prioritizes maximizing the sample size (N), which is critical for ensuring the statistical robustness of rare, long-duration trends ($d \geq 7$) that are central to the tail analysis.

Table 1: Descriptive summary of data and trends by index.

Index	Start Date	End Date	Trading Days	Total Trends	Up Trends	Down Trends
DJI	1992-01-02	2025-09-05	8480	4346	2173	2173
Nasdaq	1971-02-05	2025-09-05	13762	6142	3071	3071
Nikkei	1970-01-15	2025-09-04	13686	6713	3357	3356
IPC	1991-11-08	2025-09-05	8472	3942	1971	1971

3.2. Multi-Scale Trend Identification Algorithm

To move beyond the analysis of daily returns ($d = 1$), we apply a *local extrema detection algorithm.* This method identifies a change in the trend direction when the price moves opposite to the prevailing trend. The formal pseudocode is presented in Algorithm 1.

Algorithm 1 Trend Inflection Point Detection

procedure GENERATETRENDS($Prices$)
  $TrendsList \leftarrow$ empty list
  $CurrentTrend \leftarrow$ Null
  for $i \leftarrow 1$ to length$(Prices) - 1$ **do**
    if $Prices[i] > Prices[i-1]$ and $CurrentTrend \neq$ 'up' **then**
      add $(i-1, Prices[i-1],$ 'up'$)$ to $TrendsList$
      $CurrentTrend \leftarrow$ 'up'
    else if $Prices[i] < Prices[i-1]$ and $CurrentTrend \neq$ 'down' **then**
      add $(i-1, Prices[i-1],$ 'down'$)$ to $TrendsList$
      $CurrentTrend \leftarrow$ 'down'
    end if
  end for
  add (last index, last price, $CurrentTrend$) to $TrendsList$
  return $TrendsList$
end procedure

The result is a segmentation of the price series into contiguous uptrends and downtrends. Figure 1 (based on the plot on p. 17) [cite: 695-713] visually illustrates how the algorithm segments the price series.

Figure 1: Visual illustration of the trend segmentation algorithm applied to a Nasdaq price series.

3.3. TReturns

Once trends are identified, we define the "Trend Return" (*TReturn*) as the total log-return from the start (P_{start}) to the end (P_{end}) of a trend:

$$TReturn = \log\left(\frac{P_{\text{end}}}{P_{\text{start}}}\right)$$

We grouped these *TReturns* into three categories for analysis:

- **AllTReturns:** The set of all *TReturns*, both positive and negative.
- **UpTReturns:** The subset of *TReturns* where $TReturn > 0$.
- **|DownTReturns|:** The absolute value of *TReturns* where $TReturn < 0$, used to compare the magnitude of losses against gains.

All *TReturns* were grouped by their duration in days (Lag) for the multiscale analysis. Furthermore, as the mode is ill-defined for continuous empirical data, it was estimated throughout this study using a *Kernel Density Estimation (KDE)* approach.

3.4. Statistical Hypothesis Tests

To rigorously validate our visual findings regarding bimodality and asymmetry, we employed a set of three complementary statistical tests for the TReturn distributions at each duration.

First, to confirm the structural transition observed in the KDE plots, we applied *Hartigan's Dip Test* [43]. This non-parametric method explicitly tests the null hypothesis (H_0) of unimodality against the alternative hypothesis of multimodality. A p-value < 0.05 allows us to statistically reject the unimodality, providing formal evidence of the emergent bimodal structure in persistent trends.

Second, to assess the symmetry between gains and losses, we utilize *Welch's t-test* [44]. Unlike the standard Student's t-test, this variant does not assume equal variances (homoscedasticity), making it appropriate for comparing samples with distinct volatilities. However, while this parametric test is standard in the literature for comparing means, we interpret its results with caution, given that its validity still relies on the assumption of normality, a condition often violated by financial data.

Consequently, to address the non-normal and leptokurtic nature of the returns, we employ the *Mann-Whitney U-Test* [45] as a more robust tool for testing asymmetry. This non-parametric test, based on ranks, determines whether the entire distribution of *UpTReturns* and *|DownTReturns|* is statistically identical, without relying on assumptions about the underlying parameters. A p-value < 0.05 in this test confirms a statistically significant asymmetry in the distribution character.

3.4.1. Wasserstein Distance (Global Asymmetry)

To rigorously quantify the magnitude of the difference between the gain and loss distributions, specifically focusing on the heavy tails, we compute the *Wasserstein Distance* (also known as Earth Mover's Distance).

We applied this metric to the aggregated multi-scale sample (combining all durations) rather than conducting a lag-by-lag analysis. This aggregation strategy is methodologically necessary to maximize the sample density in the extreme tails of the distribution. Since extreme market moves are rare events, individual lag distributions often lack sufficient statistical power to allow for a robust structural comparison of tail behavior. Aggregating the data allows us to capture the global structural asymmetry of market dynamics.

For one-dimensional probability distributions, the 1^{st} Wasserstein distance between two distributions U (*UpTReturns*) and D (|*DownTReturns*|) with CDFs F_U and F_D respectively, can be explicitly defined as the L_1 norm of the difference between their CDFs:

$$W_1(F_U, F_D) = \int_{-\infty}^{\infty} |F_U(x) - F_D(x)|\, dx \tag{1}$$

Intuitively, this metric corresponds to the area between the two survival curves (1-CDF). It represents the minimum "work" required to transform the distribution of gains into the distribution of losses. This property makes it particularly suitable for our study, as it quantifies the total *Asymmetry of Character* (magnitude and volatility) across the entire domain rather than just identifying a point of statistical divergence.

Unlike the Kolmogorov-Smirnov test, which considers only the maximum local divergence, the Wasserstein metric captures the global structural differences. The statistical significance for the W_1 statistic is assessed via a *permutation test* with $N = 10,000$ iterations [46, 47], providing a robust p-value without assuming parametric forms.

3.5. Statistical Error Estimation

To quantify the uncertainty in our calculated statistics (mean, median, standard deviation, skewness, and kurtosis) at each (lag), we employed a *non-parametric bootstrapping* [48].

For each statistic at each duration d, we generated $N = 10,000$ resamples with replacement from the available empirical data. The standard error of the statistic (represented visually as error bands in our plots) is calculated as the standard deviation of the distribution of statistics obtained from these N bootstrap samples.

3.6. Monte Carlo Null Model Benchmark

To rigorously distinguish between emerging market properties and potential statistical artifacts of the trend identification algorithm, we construct a "Null Model" using a Monte Carlo simulation of a Geometric Brownian Motion (GBM)[23]. As established in recent comparative literature, GBM remains the foundational stochastic model for representing asset dynamics with constant volatility and drift, serving as the essential baseline against which complex behaviors, such as jumps and stochastic volatility, must be measured [14].

3.6.1. Model Definition and Calibration

GBM postulates that stock prices follow a continuous-time stochastic process satisfying the following stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{2}$$

where μ is the drift, σ is the volatility, and W_t is a standard Brownian motion. For our discrete-time simulation, we utilize the exact solution to this SDE, which is derived by applying *Itô's Lemma* to $f(S_t) = \log(S_t)$ [23, 2]:

$$S_t = S_{t-1} \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z_t\right] \tag{3}$$

where $Z_t \sim \mathcal{N}(0, 1)$ is a standard normal random variable.

We generated an ensemble of $M = 1000$ independent price trajectories, each calibrated to mirror the statistical depth and first-order moments of the empirical indices:

- **Duration:** $N = 8,000$ trading days (approx. 30 years), ensuring sufficient sample size to capture rare long-duration trends ($d \geq 7$).
- **Parameters:** We set an annual drift $\mu = 0.08$ and volatility $\sigma = 0.15$, consistent with historical averages for developed equity markets.
- **Time Step:** $\Delta t = 1/252$ (daily steps).

The inclusion of a positive drift ($\mu > 0$) is critical. This allows us to test whether the observed "Asymmetry of Frequency" (persistence of uptrends) is merely a mechanical consequence of a trending market—which GBM should reproduce—or a sign of endogenous memory. Conversely, because GBM assumes constant volatility, it serves as a strict control for our "Asymmetry of Character" hypothesis; any deviation from a volatility ratio of 1.0 in the empirical data would signal structural non-normality.

3.6.2. Ensemble Statistics Methodology

Unlike the empirical analysis, where confidence intervals are derived via bootstrapping a single historical realization, for the Null Model, we compute the **ensemble statistics**. For each metric (e.g., Volatility Ratio at Lag d), we calculate the value for each of the $M = 100$ independent simulations and report the mean and standard deviation across the ensemble. This provides a robust "zone of normality" for a random walk. The procedure is formalized in Algorithm 2.

Algorithm 2 Monte Carlo Ensemble Benchmark

procedure RunMonteCarlo(M, μ, σ, N)
  $EnsembleStats \leftarrow$ empty list
  for $k \leftarrow 1$ to M **do**
    $Prices \leftarrow$ GenerateGBM(μ, σ, N) ▷ Eq. 2
    $TReturns \leftarrow$ GenerateTReturns($Prices$) ▷ Apply Algo 1
    $Stats_k \leftarrow$ CalculateRatiosAndProbabilities($Trends$)
    Append $Stats_k$ to $EnsembleStats$
  end for
  $MeanBenchmark \leftarrow$ Mean($EnsembleStats$)
  $StdBenchmark \leftarrow$ StdDev($EnsembleStats$)
  return $MeanBenchmark, StdBenchmark$
end procedure

4. Results

This section presents the empirical findings of the multiscale analysis. We follow a logical progression to disentangle the structural properties of the market trends. First, we describe the basic anatomy of the trend durations. Second, we present a formal statistical description of the emergent bimodality obtained by filtering *TReturns* based on duration. Third, we analyze the multiscale asymmetry between the gain and loss distributions. Fourth, we quantify global asymmetry in the distribution tails. Finally, we contrast these findings with those of a Monte Carlo Null Model (GBM). This comparison serves a dual purpose: on one hand, it evaluates how well the GBM captures the market's multiscale trend structure, thereby highlighting the limits and potential risks of relying on classical models in quantitative finance; on the other hand, it allows us to rigorously delimit and characterize the true nature of market dynamics by isolating genuine structural properties from model artifacts.

4.1. Anatomy of Market Trends

Before analyzing the returns, we examine the underlying structure of the trend durations. **Figure 2** displays the distribution of trend lengths (in days) for all four indices on a log-frequency scale.

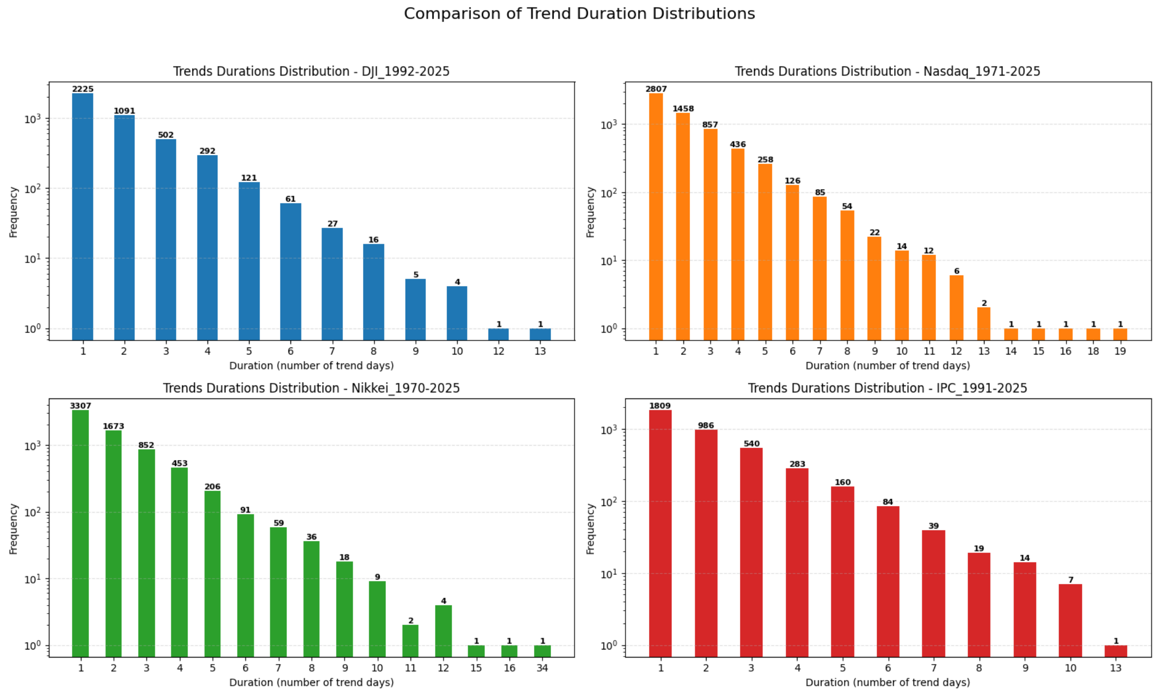

Figure 2: Trend Duration Distributions for all four indices. The vertical axis (frequency) is on a logarithmic scale, so we can observe the geometric decrease.

The distributions are consistent and unambiguous across all markets, and the vast majority of trends are short-lived. For all indices, 1-day trends were the most common, followed by a rapid geometric decay in frequency as the duration increased, which is consistent with previous empirical studies [20].

This result supports the relevance of our multiscale approach to the problem. Analyzing only 1-day returns (the mode of the distribution) effectively ignores the complex dynamics present in trends that persist for two, three, or more days. By categorizing returns based on their "intrinsic time" (duration), we open a complementary analytical window that reveals structures invisible to fixed-time analysis.

4.2. Emergent Bimodality in Trend Returns

This emergence was recently discovered and pointed out by [19]. Figure 3 presents the Kernel Density Estimation (KDE) plots for *TReturns* grouped by duration.

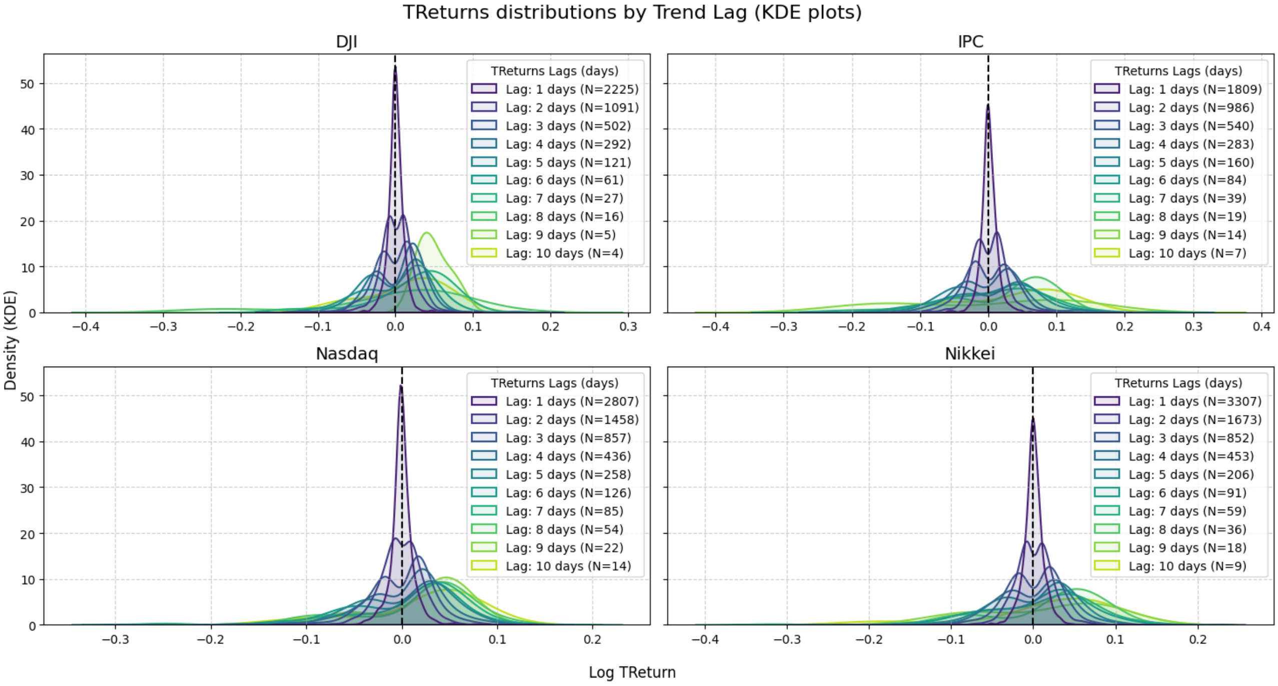

Figure 3: KDE plots of *TReturn* distributions for different time lags. Note the transition from a uni-modal peak at zero for Lag 1 to bimodal distributions for Lags ≥ 2.

A distinct evolutionary pattern was observed across time scales. While the distribution for Lag 1 ($d = 1$) is sharply unimodal and leptokurtic—representing high-frequency market noise or mean reversion—the structure fundamentally changes for Lags ≥ 2. The peak at zero vanishes and is replaced by a *dip*, separating into two distinct modes (positive and negative). Nevertheless, at some point for the longest durations, visual inspection suggests a tertiary transition back toward a "displaced unimodality," where the negative mode diminishes and the probability mass concentrates on the positive side.

The origin of this transition and the consequent disappearance of probability mass at zero for $d \geq 2$ is, to a large extent, a mechanical consequence of the trend definition: a trend persisting for multiple days implies a cumulative price movement away from the origin (i.e., the probability of a zero return for a multi-day trend is negligible). However, the emergence of a bimodal shape—rather than a single displaced mode—requires the structural coexistence of both uptrends and downtrends at that specific scale. If a market is dominated solely by long-term uptrends, the distribution becomes positively unimodal. The persistence of two distinct modes thus suggests a symmetry of existence in the market structure at intermediate scales.

To validate the extent to which bimodality persists statistically, we apply Hartigan's Dip Test (the null hypothesis establishes H_0: unimodality). As shown in Figure 4 (detailed tables in the Appendix), the p-values for Lags 2 through 5 are consistently below the $\alpha = 0.05$ significance level for all indices.

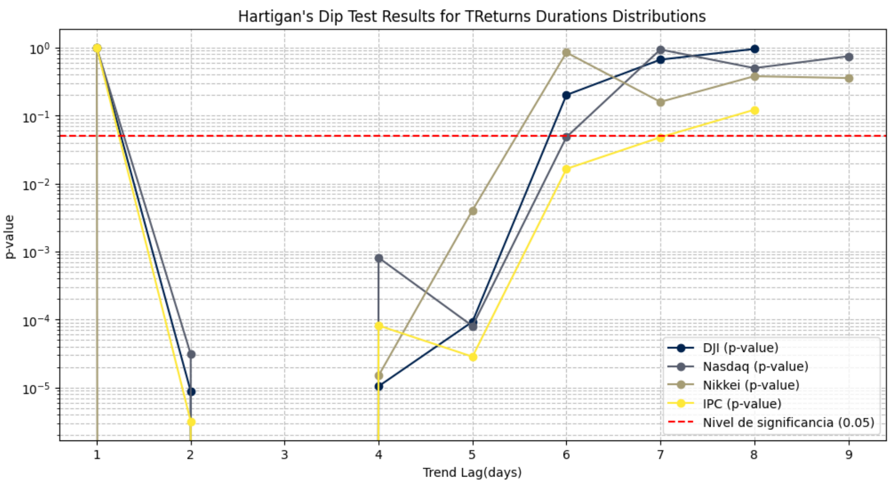

Figure 4: Hartigan's Dip Test p-values (log-scale) vs. Trend Lag. The red dashed line is the 0.05 significance level.

We thus statistically reject the null hypothesis of uni-modality consistently for trend durations between $d \geq 2$ and $d < 6$. Note that for longer lags ($d > 5$), the p-values rise. This creates a dichotomy in interpretation. This does not necessarily imply that bimodality vanishes; it could simply reflect the geometric decay of the sample size noted in Section 4.1 (e.g., N drops from thousands to double digits), which drastically reduces the statistical power of the tests. However, the uniform convergence toward a positively displaced unimodality across all four indices suggests that this is an intrinsic property of the market rather than a random artifact of small samples. If this transition is merely noise, we would expect divergent behavior. Instead, the coherent extinction of the negative mode points to structural deviation. As we demonstrate in Section 4.5, this behavior stands in sharp contrast to a random walk (GBM), which maintains a persistent bimodal structure across all lags. This implies that the "extinction" of long-term downtrends observed here is a unique signature of empirical market dynamics that is distinct from simple stochastic motion.

4.3. Multi-Scale Gain-Loss Asymmetry

Having characterized the structural evolution of trend returns, where bimodality emerges distinctly for intermediate durations ($2 \leq d \leq 5$) before showing signs of transition toward a positive displaced unimodality at longer lags, we next investigate the symmetry between these modes. We explore a dual asymmetry in terms of both frequency and character (i.e., magnitude and volatility) to help explain this structural evolution.

First, we examine the frequency of gains versus losses. Figure 5 displays the probability of a positive *TReturn* (purple) versus a non-positive one (orange).

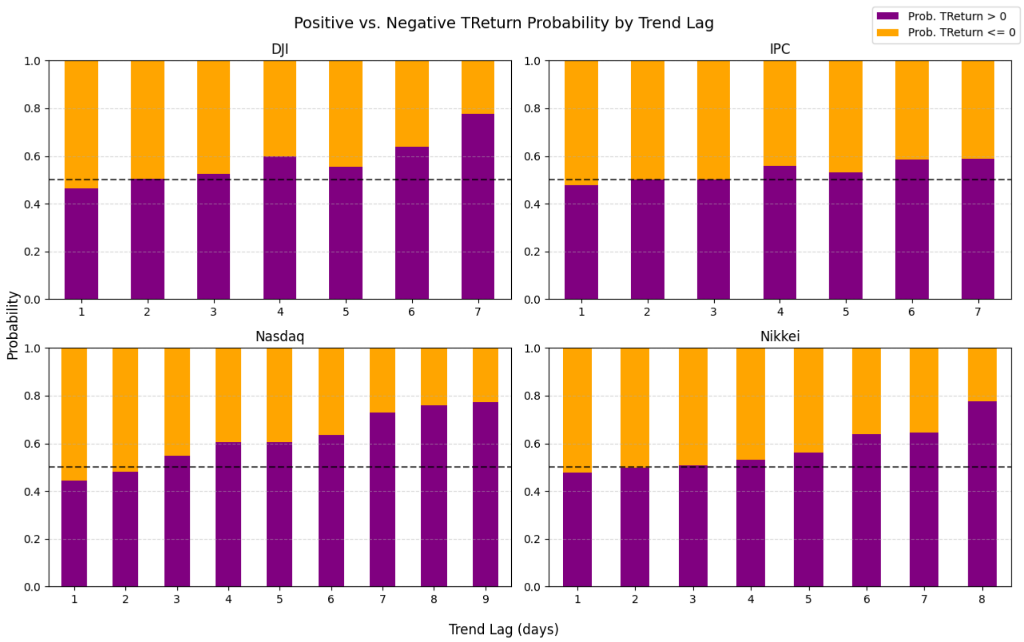

Figure 5: Positive vs. Negative *TReturn* Probability by Trend Lag.

The plots reveal a critical asymmetry that serves as a key component. While for short lags ($d = 1, 2$), the probabilities are clustered near the 50% line, for $d \geq 3$ a clear divergence emerges. For the DJI, Nasdaq, and Nikkei, the probability of a trend being an uptrend increases with duration. This suggests that the longer a trend persists, the more likely it is to be positive. This frequency asymmetry offers a mechanistic explanation for the transition observed in Section 4.2: as the duration increases, the "negative mode" effectively fades due to the lower survival rate of downtrends, leaving the "positive mode" as the dominant survivor, which creates the appearance of a displaced unimodality.

The IPC index serves as a notable exception, showing probabilities that remain stable around the 50% line. This suggests a different survival dynamic in this market, where downtrends are as resilient as the uptrends. This unique behavior provides a compelling cross-validation of our structural hypothesis: since the IPC maintains a balance between up and down trends (the negative mode is not extinguished), the bimodal structure should logically persist longer. Indeed, revisiting Figure 4 (Hartigan's test), we observe that the IPC exhibits a robust statistical rejection of unimodality that persists significantly, precisely because the structural "tug-of-war" between modes remains active.

Second, we analyze the character of these trends by plotting the ratio of *UpTReturns* statistics to *|DownTReturns|* statistics for all four indices (Figure 6). The red dashed line at Ratio = 1.0 represents perfect symmetry.

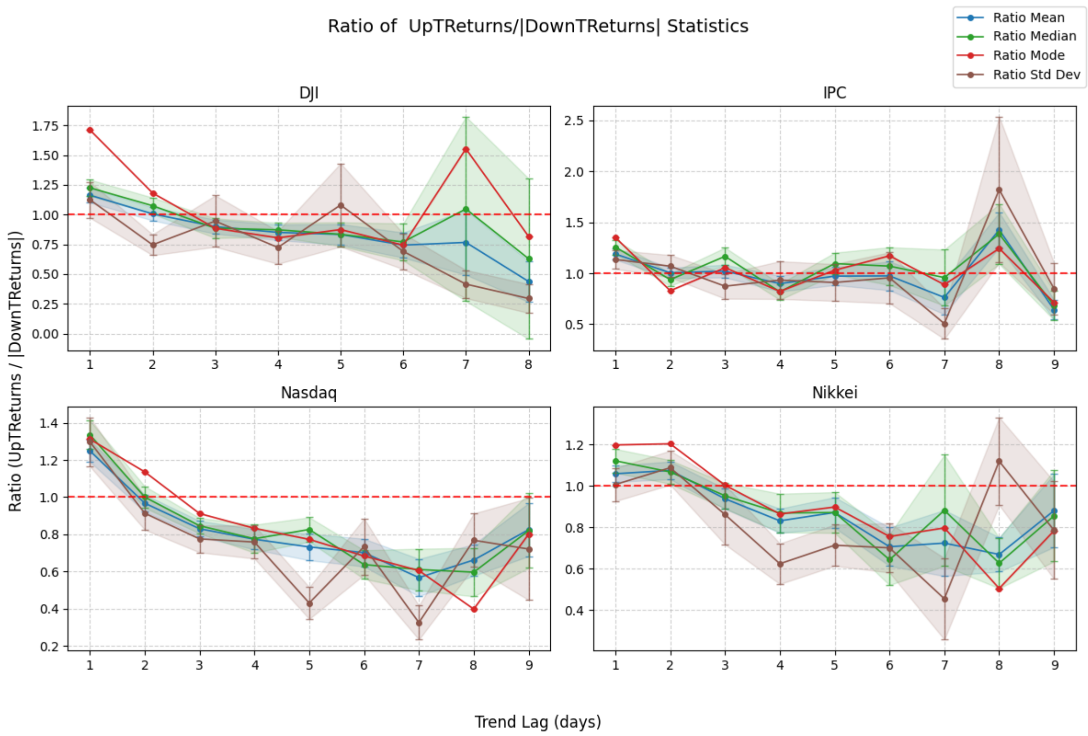

Figure 6: Ratio of UpTReturns and |DownTReturns| Statistics for all four indices. Ratios below 1.0 indicate that downtrends are structurally larger or more volatile.

This analysis revealed a counterweight to the frequency finding. While long downtrends are becoming rarer (as shown by the frequency asymmetry in developed markets), the Volatility Ratio (brown line) confirms that they remain structurally more violent. The ratio dips and stays consistently below 1.0 for $d \geq 2$ across all four indices, indicating that downtrends are structurally more volatile than uptrends. Regarding the magnitude (Mean and Median ratios), the Nasdaq and Nikkei show a consistent ratio below 1.0. The IPC exhibits a deviation at longer lags ($d \geq 8$); however, as indicated by the widened error bands in Figure 6, these estimates suffer from high uncertainty due to the small sample size ($N < 20$).

4.3.1. Asymmetry Test for multiscale analysis

To examine the statistical significance of these observations, we employed two complementary hypothesis tests, as shown in Figure 7. While standard in the literature, their application here requires a nuanced interpretation, given the structural properties of multi-scale returns.

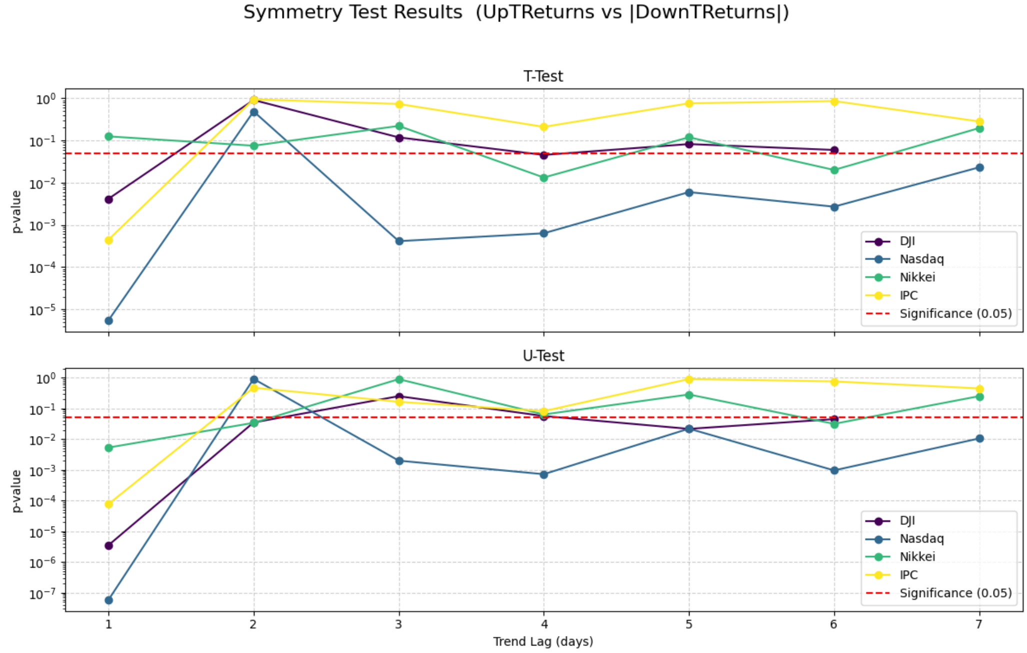

Figure 7: Symmetry Test Results (p-values, log-scale). The U-Test (bottom) provides a more robust rejection of symmetry (H_0) given the non-normal data.

First, Welch's t-test (top panel) compared the means. It shows the rejection of symmetry for several specific lags (e.g., Nasdaq lags 3–7). However, its reliability is strictly constrained by the assumption of normality—a hypothesis we explicitly rejected in Section 4.2 via the Hartigan Dip Test for $d \geq 2$. Thus, while informative, it likely provides an incomplete picture of divergence.

Second, we applied the Mann-Whitney U-Test (bottom panel). As a non-parametric test based on ranks, it is robust to outliers and does not assume a normal distribution. The results indicate a more widespread rejection of symmetry across the four markets, particularly at shorter lags ($d = 1$) and intermediate durations. However, the rejection is not universal across all lags, and for some durations, the null hypothesis of identical distributions cannot be rejected, despite the visual differences observed in the KDEs.

This intermittency is not a contradiction but rather a diagnostic signal. Because the U-Test is primarily sensitive to differences in location (median) and stochastic dominance, its variable power suggests that asymmetry is not primarily driven by a simple directional shift in central tendency. If one distribution were simply "shifted" relative to the other, the U-Test would reject consistently. Instead, the fact that location tests struggle while volatility ratios are clear confirms that divergence is structural. The "Asymmetry of Character" is rooted in the differing shapes—specifically the higher volatility and heavier tails of downtrends—rather than just their average values. This highlights why relying solely on central tendency metrics is insufficient and why analyzing the tails (via Wasserstein Distance) is essential to capture the full picture of asymmetry.

These two findings describe a market where uptrends are favored by *persistence* (frequency), whereas downtrends are defined by *intensity* (character).

4.4. Global Multiscale Asymmetry and Tail Analysis

To integrate the insights obtained from the lag-by-lag analysis, we ultimately assessed the extremities of the combined multi-scale sample. In Section 4.3, the rank-based tests (U-Test) yielded mixed results for some markets, particularly the IPC, despite the clear signal from the Volatility Ratios. This suggests that asymmetry might not be driven by a simple shift in the central tendency but by the structural shape of the tails. Aggregating the data allows us to map this anatomy with maximum resolution, quantifying the well-known stylized fact of gain-loss asymmetry through a multi-scale lens.

Figure 8 presents the Survival Plots (1-CDF) on a logarithmic scale. This visualization highlights the contrast in survival dynamics. While Section 4.3 demonstrated that uptrends tend to survive longer in *time* (duration), the survival plots reveal that downtrends (orange curve) consistently track above uptrends (purple curve) in magnitude. This indicates that negative trends have a higher probability of causing extreme price changes. In other words, while gains persist for a longer period, losses travel further.

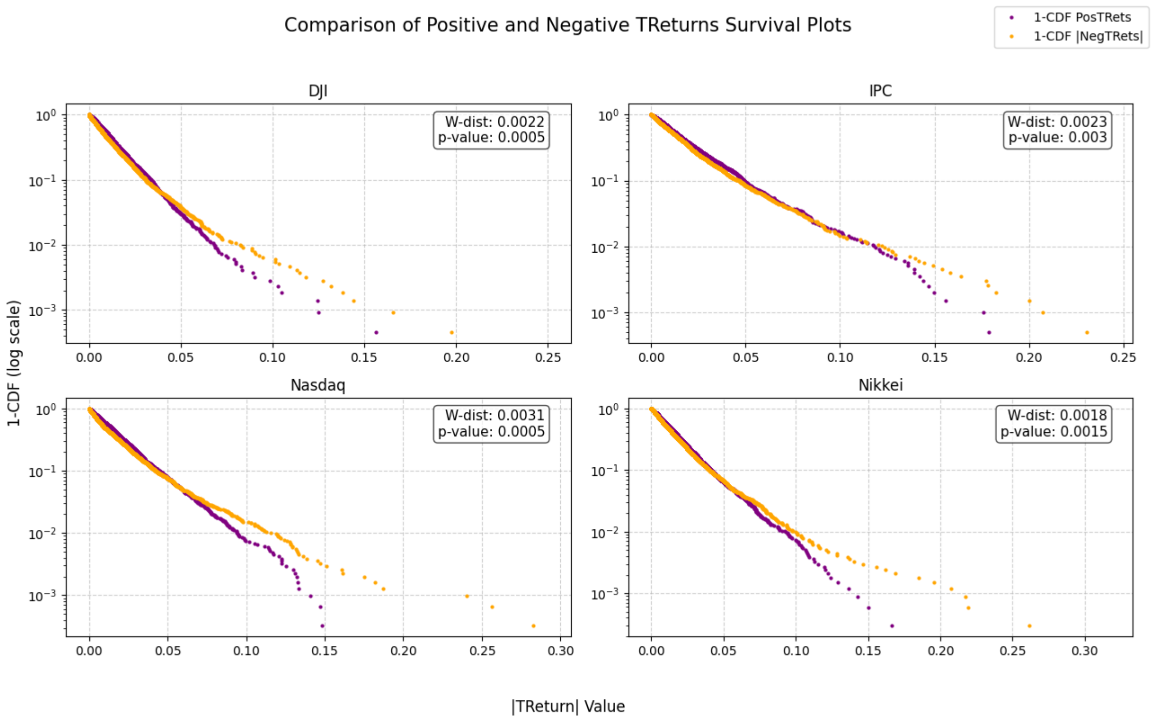

Figure 8: Survival plots (1-CDF) for UpTReturns vs |DownTReturns|. The Wasserstein distance confirms asymmetry.

To rigorously quantify this structural gap, we compute the Wasserstein Distance (W_1), which measures the total "work" or area between the two curves. The results offer a precise characterization of asymmetry across markets. For developed markets such as the DJI ($W = 0.0022$) and Nasdaq ($W = 0.0031$), the metric confirms a clear systemic divergence. Interestingly, the IPC yields a high Wasserstein distance ($W = 0.0023$), surpassing the Nikkei ($W = 0.0018$), despite the visual proximity of its survival curves in the probability plot. This resolves the anomaly observed in Section 4.3: while the IPC is symmetric in *frequency*, it is highly asymmetric in *magnitude* at the extreme tails. The Wasserstein metric captures the heavy "cost" of those rare but violent corrections typical of emerging markets, which are less apparent in central tendency metrics.

In practical terms, these values imply that downtrends carry an inherent structural penalty—an "excess magnitude" of approximately 0.2

4.5. Validation against Null Model; Revealing the Limits of GBM

To distinguish between the genuine structural properties of the market and statistical artifacts inherent to the trend identification algorithm, we contrast our findings with the Monte Carlo Null Model (GBM) described in Section 3.6. This comparison serves as a rigorous control group, isolating the mechanical consequences from the true market dynamics.

Figure 9 presents the comprehensive results of the simulation compared with the stylized facts observed in the empirical data. This comparison reveals a clear hierarchy of market properties. First, regarding Bimodality, Panel A confirms that the GBM reproduces the bimodal distribution for intermediate durations. However, a crucial divergence appears at longer lags. While empirical markets transition toward a displaced unimodality for $d > 6$ (likely due to the extinction of downtrends), the GBM maintains a persistent bimodal structure across all simulated lags. This suggests that the loss of the negative mode in real markets is a non-trivial feature of market exhaustion that is not present in a random walk.

Second, regarding **Frequency**, Panel B confirms that a random walk with positive drift inherently increases the probability of uptrends with duration. However, a crucial divergence emerges in formal statistical tests. As shown in Table B.3 (Appendix), the GBM consistently fails to reject the null hypothesis of symmetry ($p > 0.05$) for the individual lags. Despite the drift, the distributions of up and down trends in the Null Model remained statistically symmetric.

This contrasts sharply with the empirical data (Figure 7), where symmetry is strongly rejected across multiple durations (e.g., Nasdaq) or, as seen in the IPC, exhibits a strong initial rejection in lag 1, followed by intermittency (Table B.3). This confirms that while the GBM remains symmetric, the market's asymmetry is structural, manifesting as intermittent statistical rejection driven by structural divergence in shape and volatility rather than just location. This implies that the "Asymmetry of Frequency" in real markets is accompanied by a fundamental structural divergence that the GBM cannot replicate: the market is not just "drifting up,"; it is structurally deforming.

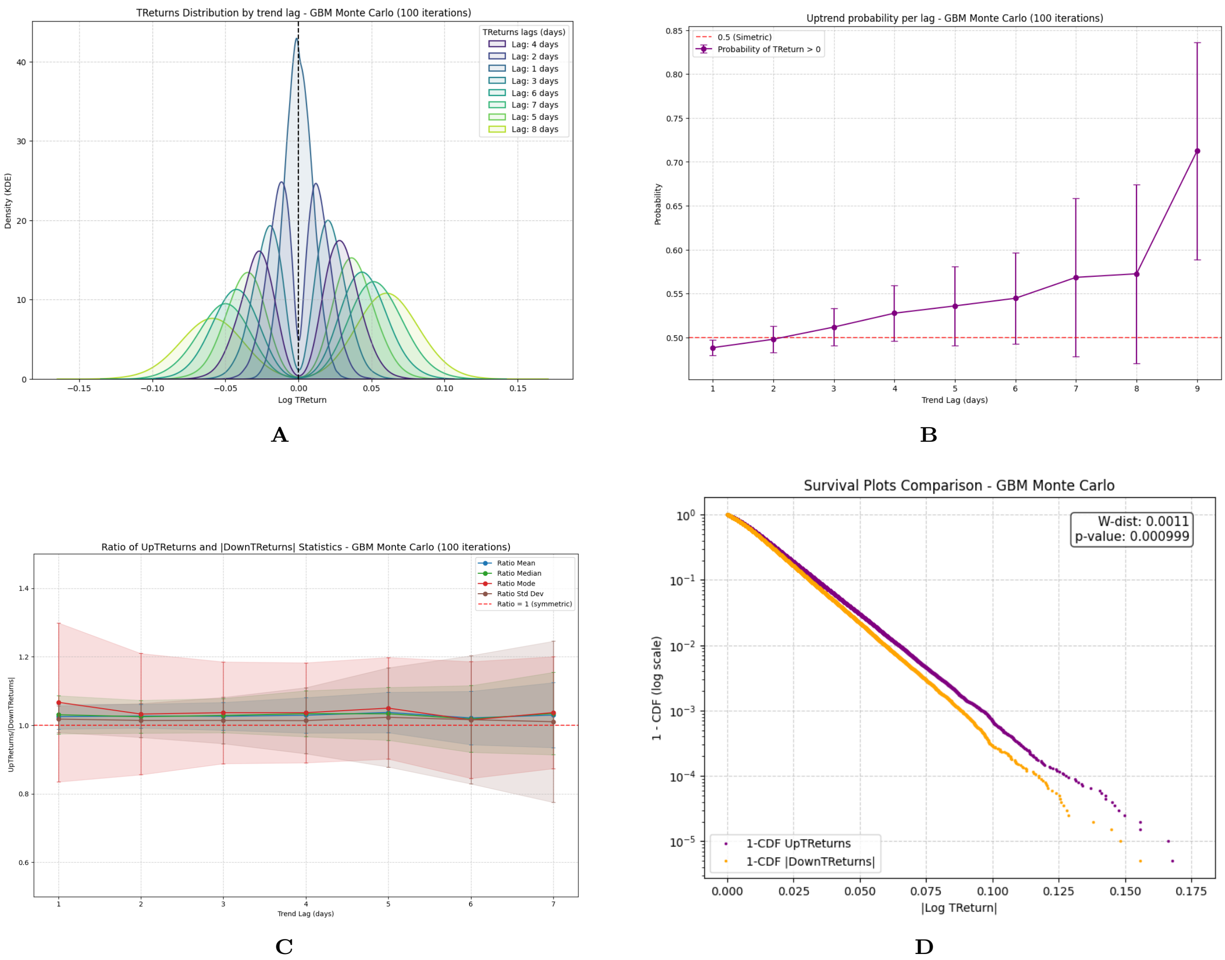

Figure 9: Monte Carlo Benchmark Results ($N = 100$ simulations). **(A)** KDE plots show emergent bimodality, confirming it is a mechanical artifact of filtering $d \geq 2$. **(B)** Up-Trend probability increases with duration, confirming that frequency asymmetry can be explained by simple positive drift ($\mu = 0.08$). **(C)** Crucially, Volatility and Magnitude Ratios remain strictly at 1.0 (dashed line), failing to reproduce the structural asymmetry of real markets. **(D)** Survival plots show symmetric tails ($W \approx 0.001$), contrasting with the heavy loss tails of empirical data.

However, the Null Model fails to reproduce the Asymmetry of Character. Panel C shows the starkest contrast. In the GBM, despite the drift, the ratios of Volatility, Mean, and Median remain strictly flat at 1.0 (within error bands). The volatility of a random uptrend is identical to that of the random downtrend. Consequently, the empirical deviations (Ratios < 1) are robust signatures of structural risk asymmetries. This is further confirmed by the Survival Plots in Panel D, where the GBM curves are nearly coincident, contrasting with the heavy loss tails of the empirical data.

To quantify this global asymmetry, Table 2 presents the statistical test results for the aggregated samples. While the GBM also rejects the null hypothesis of symmetry ($p < 0.01$), this rejection is driven primarily by the location shift caused by the drift. In contrast, the rejection in empirical markets (with p-values as low as 10^{-17}) is driven by the structural divergence in shape and volatility identified above.

Table 2: Global Symmetry Test Results for All Trend Durations Combined.

	Welch's T-Test (Means)		**Mann-Whitney U-Test (Distributions)**	
Index	**t-statistic**	**p-value**	**U-statistic**	**p-value**
DJI	2.4677	0.01364	2,628,473.0	9.928e-11
Nasdaq	3.1240	0.00179	5,307,886.0	1.524e-17
Nikkei	1.7360	0.08261	6,044,014.0	2.263e-07
IPC	2.4970	0.01257	2,101,857.0	8.101e-06
GBM	3.7383	1.874e-04	3,256,168.0	8.777e-03

Finally, Figure 10 compares the aggregate statistics across all the lags. The analysis of central tendency metrics revealed a persistent positive drift for both the GBM and empirical indices. However, the Skewness panel highlights a critical divergence: while the GBM skewness fluctuates near zero, all real markets consistently drop into deep negative territory for $d \geq 2$, confirming that "Universal Negative Skewness" is a non-random structural property. Regarding Kurtosis, the behavior is fundamentally distinct. The GBM starts and remains stable near zero (consistent with Gaussian behavior), whereas empirical indices exhibit extreme leptokurtosis at $d = 1$ followed by rapid decay. This confirms that the "Gaussianization" of tails is a convergence process unique to the empirical data as trends persist, in contrast to the GBM, which is Gaussian by design at all scales.

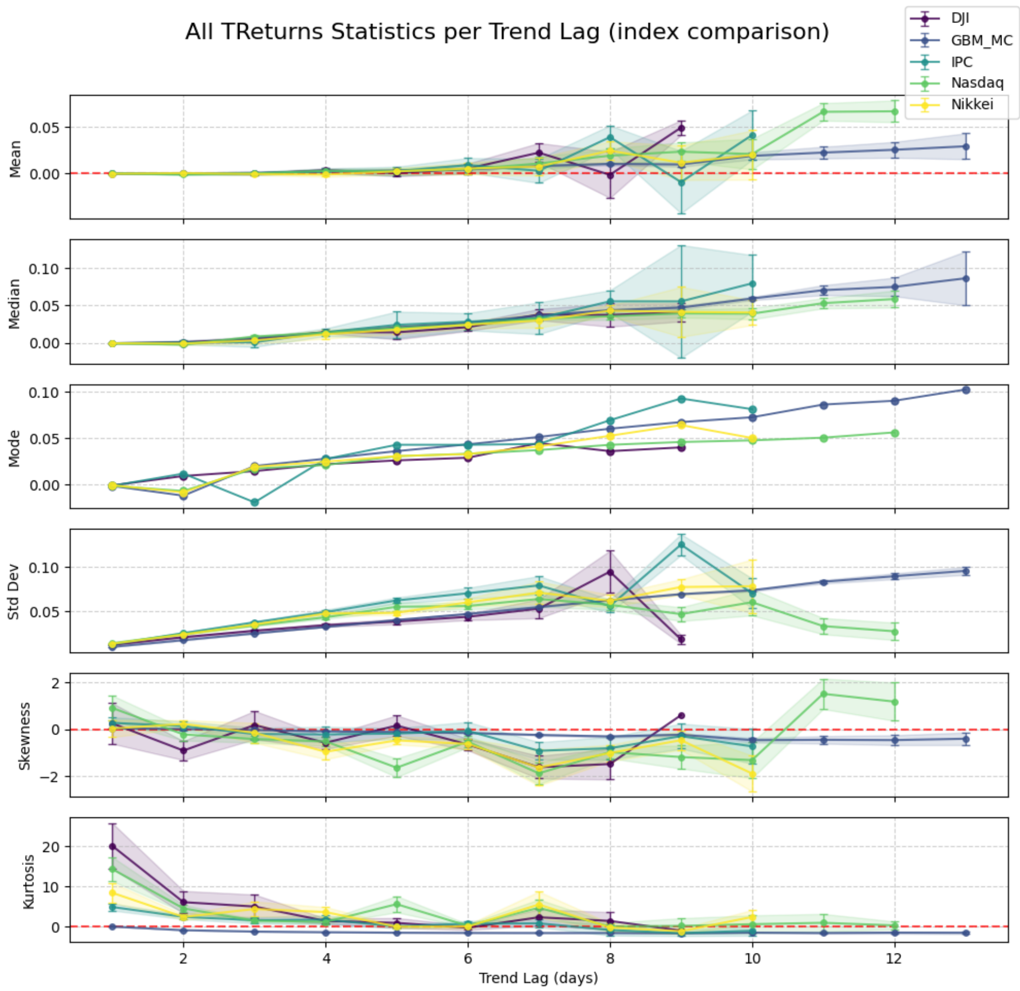

Figure 10: Comparison of AllTReturns statistics: Empirical Indices vs. GBM Benchmark. Note the critical divergence in skewness (Panel 5): while the GBM (grey line) remains symmetric at zero, all real markets (colored lines) drop significantly into the negative territory. Note also the distinction in Kurtosis (Panel 6): empirical markets start with high leptokurtosis and decay, whereas GBM remains stable near zero.

5. Discussion and Conclusions

Based on our empirical findings, rigorously validated against a Monte Carlo Null Model, we draw the following conclusions about the theoretical foundations and practical applications in financial modeling. Our analysis enhances the characterization of market behavior by distinguishing mechanically induced trend continuation effects from genuine structural irregularities.

5.1. The Nature of Bimodality in Markets

We document that the emergence of bimodality is largely a mechanical consequence of filtering trends through persistence. As confirmed by our benchmark, even a Geometric Brownian Motion (GBM) exhibits this separation when daily noise ($d = 1$) is removed. However, a critical structural divergence appears at longer timescales. While the GBM maintains a persistent bimodal structure across all simulated lags, empirical markets exhibit a transition toward a shifted unimodality at long durations ($d > 6$).This phenomenon represents the "extinction" of the negative mode, revealing that real-world downtrends face a significantly higher mortality rate than random walks. Crucially, this topological transition is the structural manifestation of the Asymmetry of Frequency: because positive trends are disproportionately more likely to survive over long horizons, the negative mode eventually starves and vanishes, leaving the positive mode as the sole survivor. This confirms that asymmetry is not just a statistical bias in counts but a fundamental dynamic limit: while random walks allow for indefinite negative runs, real markets enforce a mean-reversion mechanism—likely driven by value buying after panic—that dismantles long-duration negative trends. Thus, the breakdown of bimodality at long scales serves as a robust stylized fact that distinguishes market dynamics from stochastic processes.

5.1.1. Empirical Substrate for Latent States

To model trend persistence and non-linearity, approaches such as Hidden Markov Models (HMM) assume that markets switch between latent "hidden" states (e.g., bull vs. bear) and rely on probabilistic inference [13]. Our multiscale analysis reveals that this dual nature is not merely a hidden state requiring blind inference but a directly observable property of the return distribution when filtered by trend duration.

While our Monte Carlo simulation confirmed that the separation into two modes was partly a mechanical consequence of filtering persistence, the empirical distributions differed fundamentally from the null model in terms of their structure. This confirms that the "two-state" hypothesis is not a statistical abstraction but reflects tangible market dynamics. Consequently, data-driven models can reduce parameter uncertainty by grounding their "latent" states in this observable bimodal reality, using the empirical moments of $d \geq 2$ trends as priors for regime calibration.

5.1.2. Theoretical Implications: Refining the Regime Hypothesis

Our findings challenge the standard specifications of the regime-switching models. Standard HMMs often assume Gaussian distributions in states. However, the contrast with the GBM (which maintains perfect symmetry) highlights that real market regimes are structurally distinct. Specifically, the negative mode is structurally more volatile and has a larger magnitude than the positive mode. This suggests that theoretical models must incorporate asymmetric emission distributions (e.g., Skewed Student-t) to accurately capture the empirically observed "violence" of the down state, which Gaussian-based HMMs fail to reproduce.

5.2. Gain-Loss Asymmetry: Theory Validation

Our multi-scale analysis reveals two distinct manifestations of asymmetry: *Asymmetry of Character* (volatility/magnitude) and *Asymmetry of Frequency* (persistence). These findings provide robust empirical validation for fundamental financial theories.

5.2.1. The Limits of Gaussian Symmetry

One of the significant theoretical contributions is the isolation of the Asymmetry of Character. While the GBM maintains perfect symmetry in volatility (Ratio ≈ 1.0) and tail behavior, empirical markets display a structural "violence" in downtrends (Ratio < 1 and high Wasserstein Distance) that the Gaussian model cannot structurally replicate [14]. This confirms that the market's deviation from normality is not merely about heavy tails in general but about a specific, asymmetric structural displacement of risk that challenges the symmetry assumptions of standard pricing models.

5.2.2. The Leverage Effect

Our finding that downtrends are intrinsically more volatile than uptrends (Ratio of Volatilities < 1) serves as a direct multi-scale confirmation of the Leverage Effect [32]. Standard literature documents that losses increase future volatility more than gains [27]. Our observation that negative trend returns accumulate larger magnitudes over shorter durations provides structural evidence for this phenomenon. While the GBM benchmark shows a volatility ratio strictly at 1.0, the empirical deviation confirms that this is a genuine market anomaly supported by "volatility feedback" mechanisms [33].

5.2.3. Behavioral Mechanics and Endogenous Instability

The empirical "violence" of downtrends strongly aligns with Behavioral Finance, particularly Prospect Theory [15]. Investors exhibit *loss aversion*, being more sensitive to losses than to equivalent gains [36]. This psychological asymmetry can trigger panic selling [33], creating a steeper marginal utility curve for losses, in contrast to the gradual and cautious accumulation of gains, often referred to as a "wall of worry."

Our results suggest that the structural *Asymmetry of Character* is the empirical footprint of this mechanism. Specifically, the significant Wasserstein Distance ($p < 0.01$) quantifies this bias in physical terms. Because the Wasserstein metric represents the minimum "work" required to transform one distribution into another, our result confirms that the market operates in a "higher energy state" during downtrends. This aligns with Sornette's theory of endogenous instability [49, 16], which posits that extreme negative regimes or crashes are not merely random outliers but critical phase transitions ("Dragon Kings") driven by positive feedback loops. Interpreting our results through this framework, the slow accumulation of gains is analogous to a buildup of potential energy, while the violent downtrends —characterized by a structural excess in magnitude quantified by our Wasserstein metric—represent the rapid, "high-energy" release of this tension. This energy signature was notably absent in the GBM control group, isolating it as a property of complex adaptive systems.

5.2.4. Momentum and Frequency: The Mechanism of Mode Extinction

Finally, the increasing probability of uptrends with duration aligns with the phenomenological spirit of Time Series Momentum (TSMOM) [39]. While TSMOM addresses the predictive power of past returns, our ex-post structural analysis reveals a consistent bias where long-duration trends are disproportionately likely to be positive.

Although our Null Model indicates that positive drift accounts for baseline frequency asymmetry, the empirical phenomenon observed here transcends simple stochastic drift. This structural persistence aligns with behavioral models of investor under-reaction [42], which suggest that information diffuses gradually during bull markets, granting uptrends a unique longevity [41].

Crucially, this asymmetry serves as the mechanistic driver of the topological transition identified in Section 4.2. The "survival bias" favoring uptrends—fueled by under-reaction—becomes so pronounced at long durations that it leads to the structural extinction of the negative mode. Unlike the GBM control, where the negative mode persists indefinitely despite the drift, real markets exhibit bounded tolerance for continuous drawdowns. This suggests that the " momentum " observed in financial time series acts as a filtering force, dismantling long-term negative structures and collapsing the distribution into a positive unimodality—a dynamic stability absent in the random walk.

5.3. Global Conclusions and Implications

The bimodality transition to shifted unimodal in market data (or extinction of negative mode for larger trends in real market data) and the consistency of the Asymmetry of Character across the DJIA, Nasdaq, Nikkei, and IPC suggest that these are robust *stylized facts of financial dynamics*, transcending specific market microstructures. Consequently, the multiscale statistics derived here—specifically, the Volatility Ratios and Wasserstein Distances—establish new, rigorous benchmarks for validating synthetic market models. Any realistic Agent-Based Model or generative algorithm must be capable of reproducing these specific asymmetries (not just the bimodal shape) to be considered a valid representation of market reality [31].

5.3.1. Theoretical Implications

We provide definitive evidence that challenges the sufficiency of the Gaussian paradigm in this regard. The can be mechanical, but the Structural Asymmetry of the modes is not. The confirmation of heavier loss tails [26] and universal negative skewness mandates that economic models explicitly incorporate asymmetric error distributions to avoid mis-specification.

5.3.2. Practical Implications: Risk Management and Model Selection

The empirical confirmation of *Asymmetry of Character* has critical implications for risk management. Traditional models that assume symmetry (including standard GBM-based VaR) systematically underestimate tail risk. By ignoring the negative skewness and the Leverage Effect manifested in our volatility ratios, these models fail to account for the structural violence of market corrections.

Consequently, our results support the transition towards advanced models. While GARCH models address volatility clustering, they often require asymmetric innovations (e.g., EGARCH, GJR-GARCH) to match reality [27]. Similarly, regime-switching models must move beyond Gaussian states to asymmetric distributions (e.g., Skewed Student-t) [14] to capture the baseline asymmetry defining market reality.

5.3.3. Future Work

This study opens avenues for the development of empirically informed Hidden Semi-Markov Models (HSMMs), where transition probabilities are calibrated using our duration-dependent probability curves. Because our data show that regime survival probabilities are duration-dependent (Asymmetry of Frequency) and regime volatilities are structurally distinct (Asymmetry of Character), future generative models should calibrate their transition matrices and emission distributions using these empirical curves rather than standard Gaussian priors. Additionally, in algorithmic trading, the clear structural boundary between unimodal noise ($d = 1$) and bimodal trends ($d \geq 2$) offers a data-driven threshold for optimizing trend-following filters, potentially reducing "whipsaw" losses by distinguishing the signal from noise based on intrinsic time.

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Appendix A. Statistical Test Formulas

Appendix A.1. Welch's t-Test (Asymmetry of Means)

This parametric test compares the means of UpTReturns ($\bar{X}_{Up}$) and |DownTReturns| ($\bar{X}_{Down}$) without assuming equal population variances. The use of this variant is critical because of the confirmed volatility asymmetry in the data.

Test Statistic and Interpretation

The t-statistic measures the difference between the means relative to the standard error of the difference.

$$t = \frac{\bar{X}_{Up} - \bar{X}_{Down}}{\sqrt{\frac{s_{Up}^2}{n_{Up}} + \frac{s_{Down}^2}{n_{Down}}}}$$

A value of $\mathbf{t < 0}$ (negative) implies that $\bar{X}_{Down}$ (the magnitude of losses) is significantly larger than $\bar{X}_{Up}$ (gains). This sign, when statistically significant, validates the *Asymmetry of Character* by its magnitude.

The degrees of freedom (ν) were adjusted using the Welch-Satterthwaite equation to account for unequal variances ($s_{Up}^2 \neq s_{Down}^2$):

$$\nu \approx \frac{\left(\frac{s_{Up}^2}{n_{Up}} + \frac{s_{Down}^2}{n_{Down}}\right)^2}{\frac{(s_{Up}^2/n_{Up})^2}{n_{Up}-1} + \frac{(s_{Down}^2/n_{Down})^2}{n_{Down}-1}}$$

Appendix A.2. Mann-Whitney U-Test (Asymmetry of Distributions by Ranks)

This non-parametric test determines whether the two entire distributions are identical, serving as the primary robust tool for asymmetry in non-normal data analysis.

The U statistic is calculated based on the sum of ranks (R_{Up}) of the *UpTReturns* sample (n_{Up}).

$$U_{Up} = n_{Up}n_{Down} + \frac{n_{Up}(n_{Up}+1)}{2} - R_{Up}$$

For large samples (n), the p-value is approximated using a Z-score as follows,

$$Z = \frac{U - \mu_U}{\sigma_U}$$

where μ_U (the expected value under symmetry) and σ_U are,

$$\mu_U = \frac{n_{Up}n_{Down}}{2}$$

$$\sigma_U = \sqrt{\frac{n_{Up}n_{Down}(n_{Up}+n_{Down}+1)}{12}}$$

The U value must be compared with the expected value under the null hypothesis of symmetry (μ_U). If U is significantly smaller than μ_U, it confirms that |DownTReturns| tends to occupy higher ranks. This signifies that losses are stochastically larger and structurally more violent than gains.

Appendix A.3. Hartigan's Dip Statistic (Test of Bimodality)

The Dip statistic (D) provides an absolute measure of evidence against unimodality. It measures the maximum distance between the observed empirical distribution (F_n) and the closest unimodal distribution (U). Higher Dip values (as observed in intermediate lags $d = 2$ to $d = 5$) provide strong quantitative evidence of the structural separation of the two modes, validating the finding of "Emergent Bimodality."

The statistic D is defined as,

$$D = \min_{U \in \mathcal{U}} \sup_x |F_n(x) - U(x)|.$$

The full analytic expression for D is derived through an algorithmic optimization process and is not typically presented as a closed-form equation in the applied literature.

Appendix B. Statistical Test Tables

Table B.3: t-test and Mann–Whitney U-test statistics by index and duration

Index	Lag	t-statistic	t-p-value	U-statistic	U-p-value
DJI	1	2.869598	0.004151	685914.0	0.000004
	2	0.113142	0.909941	159746.0	0.034922
	3	-1.563801	0.118513	29551.0	0.250601
	4	-2.012912	0.045498	8889.0	0.056587
	5	-1.752564	0.082304	1368.0	0.021630
	6	-1.951626	0.059702	295.0	0.044954
GBM	1	1.067660	0.285777	804025.0	0.326063
	2	-0.025044	0.980023	210451.0	0.961258
	3	0.661936	0.508280	42666.0	0.407918
	4	1.157810	0.247912	11125.0	0.229854
	5	-1.115052	0.266583	3066.0	0.338226
	6	-1.667458	0.099821	697.0	0.149988
	7	1.020412	0.314070	243.0	0.407570
IPC	1	3.518371	0.000445	452148.0	0.000077
	2	0.070733	0.943624	118315.0	0.472998
	3	0.337682	0.735737	38971.0	0.164432
	4	-1.254924	0.210646	8685.0	0.081885
	5	-0.305671	0.760277	3152.0	0.904740
	6	-0.174533	0.861942	892.0	0.757717
	7	-1.102145	0.283313	157.0	0.449273
Nasdaq	1	4.555965	0.000005	1088340.0	0.00000006
	2	-0.692058	0.489011	266139.0	0.913937
	3	-3.548762	0.000412	79800.0	0.001999
	4	-3.453938	0.000633	18357.0	0.000725
	5	-2.796343	0.005982	6613.0	0.021973
	6	-3.105770	0.002703	1188.0	0.000963
	7	-2.428103	0.023152	454.0	0.010556
Nikkei	1	1.533348	0.125287	1440968.0	0.005333
	2	1.780228	0.075222	370810.0	0.033970
	3	-1.223207	0.221602	90240.0	0.896860
	4	-2.489360	0.013269	22967.0	0.063651
	5	-1.566402	0.119303	4765.0	0.284170
	6	-2.402475	0.020047	696.0	0.031518
	7	-1.326693	0.196828	326.0	0.251067

Table B.4: Hartigan's Dip Test statistics and P-Values by Trend Duration (d). Testing the Null Hypothesis of Unimodality (H_0). Values < 0.05 (bold) indicate a statistically significant bimodality. Note the divergence at longer lags ($d \geq 6$), where empirical markets lose bimodality (p-values rise) while the GBM maintains significant bimodality ($p < 0.01$) for almost all simulated durations.

duration	**DJI**		**GBM**		**Nasdaq**		**Nikkei**		**IPC**	
	Dip	p-value	Dip	p-value	Dip	p-value	Dip	p-value	Dip	p-value
1	0.003553	1.000000	0.003895	0.996975	0.003914	0.995165	0.003424	0.996697	0.003245	1.000000
2	0.027488	**0.000009**	0.040583	**0.000000**	0.022425	**0.000031**	0.024995	**0.000000**	0.030631	**0.000003**
3	0.044436	**0.000000**	0.073512	**0.000000**	0.035432	**0.000000**	0.038934	**0.000000**	0.044624	**0.000000**
4	0.050833	**0.000010**	0.091006	**0.000000**	0.034647	**0.000812**	0.041098	**0.000015**	0.047668	**0.000082**
5	0.070142	**0.000093**	0.088563	**0.000000**	0.049889	**0.000080**	0.045191	**0.004048**	0.064913	**0.000028**
6	0.053576	0.199994	0.106432	**0.000000**	0.046042	**0.048085**	0.030047	0.841992	0.061601	**0.016398**
7	0.058961	0.661575	0.115753	**0.000040**	0.028276	0.932079	0.056767	0.158284	0.078855	**0.047810**
8	0.059035	0.950086	0.106942	**0.007559**	0.047185	0.497084	0.060557	0.377599	0.097423	0.121297
9	-	-	-		0.061586	0.742501	0.083478	0.354949	-	-

Table B.5: Estadísticas multiescala de AllTReturns para DJI e IPC. Cada entrada es estimador ± error estándar (4 decimales).

Index	duration	mean ± err	median ± err	std dev ± err	skewness ± err	kurtosis ± err
DJI	1	0.0000 ± 0.0003	-0.0004 ± 0.0001	0.0119 ± 0.0006	0.2525 ± 0.8827	20.0604 ± 5.7536
	2	0.0002 ± 0.0006	0.0011 ± 0.0017	0.0204 ± 0.0009	-0.9086 ± 0.3995	6.0574 ± 2.8066
	3	-0.0000 ± 0.0013	0.0051 ± 0.0035	0.0277 ± 0.0016	0.1839 ± 0.5717	4.9640 ± 3.0642
	4	0.0034 ± 0.0020	0.0146 ± 0.0014	0.0341 ± 0.0018	-0.5740 ± 0.2647	1.3810 ± 0.5892
	5	0.0005 ± 0.0034	0.0142 ± 0.0094	0.0382 ± 0.0028	0.1626 ± 0.4402	0.8883 ± 1.2255
	6	0.0053 ± 0.0056	0.0211 ± 0.0057	0.0436 ± 0.0037	-0.6467 ± 0.2511	-0.3101 ± 0.5201
	7	0.0224 ± 0.0100	0.0379 ± 0.0071	0.0528 ± 0.0110	-1.6189 ± 0.4853	2.2926 ± 2.0161
	8	-0.0013 ± 0.0236	0.0373 ± 0.0148	0.0947 ± 0.0236	-1.4841 ± 0.6484	1.3815 ± 2.3149
	9	0.0491 ± 0.0080	0.0409 ± 0.0123	0.0180 ± 0.0050	0.6053 ±	-1.0132 ± 0.6697
IPC	1	0.0004 ± 0.0003	-0.0004 ± 0.0002	0.0123 ± 0.0004	0.2736 ± 0.2387	4.8456 ± 0.8881
	2	0.0001 ± 0.0008	0.0007 ± 0.0027	0.0251 ± 0.0009	0.1406 ± 0.2044	2.3975 ± 0.5367
	3	0.0003 ± 0.0017	0.0011 ± 0.0062	0.0372 ± 0.0015	-0.1998 ± 0.2476	1.5562 ± 0.7382
	4	0.0025 ± 0.0029	0.0144 ± 0.0047	0.0489 ± 0.0027	-0.2166 ± 0.3404	1.7028 ± 1.0792
	5	0.0026 ± 0.0048	0.0242 ± 0.0163	0.0621 ± 0.0035	-0.1797 ± 0.2336	0.0355 ± 0.4705
	6	0.0090 ± 0.0078	0.0284 ± 0.0118	0.0702 ± 0.0063	-0.0637 ± 0.3746	0.6864 ± 0.7193
	7	0.0030 ± 0.0123	0.0330 ± 0.0217	0.0791 ± 0.0109	-0.9180 ± 0.3545	0.7914 ± 0.9009
	8	0.0390 ± 0.0127	0.0555 ± 0.0139	0.0557 ± 0.0075	-0.8074 ± 0.4982	-0.8795 ± 1.4688
	9	-0.0091 ± 0.0335	0.0554 ± 0.0760	0.1254 ± 0.0120	-0.2924 ± 0.5807	-1.6488 ± 0.7665
	10	0.0412 ± 0.0264	0.0794 ± 0.0397	0.0702 ± 0.0166	-0.7260 ± 0.7697	-1.0501 ± 1.2704

Table B.6: Estadísticas multiescala de AllTReturns para Nasdaq y Nikkei. Cada entrada es estimador ± error estándar (4 decimales).

Index	duration	mean ± err	median ± err	std dev ± err	skewness ± err	kurtosis ± err
Nasdaq	1	-0.0000 ± 0.0003	-0.0007 ± 0.0001	0.0136 ± 0.0005	0.9143 ± 0.5243	14.2864 ± 3.0014
	2	-0.0010 ± 0.0006	-0.0020 ± 0.0010	0.0240 ± 0.0008	-0.2173 ± 0.2613	4.5001 ± 0.8104
	3	0.0001 ± 0.0012	0.0085 ± 0.0015	0.0338 ± 0.0010	-0.4150 ± 0.1563	1.4326 ± 0.3424
	4	0.0030 ± 0.0020	0.0138 ± 0.0015	0.0430 ± 0.0018	-0.5138 ± 0.1816	1.1425 ± 0.4845
	5	0.0024 ± 0.0034	0.0197 ± 0.0020	0.0549 ± 0.0046	-1.6413 ± 0.3931	5.5576 ± 1.9539
	6	0.0047 ± 0.0049	0.0246 ± 0.0031	0.0558 ± 0.0037	-0.5247 ± 0.2433	0.1716 ± 0.4120
	7	0.0107 ± 0.0071	0.0325 ± 0.0020	0.0637 ± 0.0088	-1.8815 ± 0.4411	4.6436 ± 1.9771
	8	0.0191 ± 0.0078	0.0356 ± 0.0047	0.0571 ± 0.0056	-0.9740 ± 0.2685	-0.0041 ± 0.8458
	9	0.0234 ± 0.0095	0.0395 ± 0.0054	0.0466 ± 0.0080	-1.1857 ± 0.5081	0.1459 ± 1.9056
	10	0.0208 ± 0.0162	0.0391 ± 0.0108	0.0600 ± 0.0140	-1.3172 ± 0.7743	0.5663 ± 2.3227
	11	0.0660 ± 0.0098	0.0530 ± 0.0068	0.0330 ± 0.0092	1.5195 ± 0.6531	0.9143 ± 2.1646
	12	0.0666 ± 0.0113	0.0585 ± 0.0120	0.0271 ± 0.0099	1.1859 ± 0.7930	0.2732 ± 0.9813
Nikkei	1	-0.0001 ± 0.0002	-0.0004 ± 0.0001	0.0131 ± 0.0004	0.0338 ± 0.3789	8.3849 ± 2.4795
	2	0.0006 ± 0.0006	-0.0007 ± 0.0018	0.0235 ± 0.0006	0.2305 ± 0.1652	2.3745 ± 0.6474
	3	-0.0004 ± 0.0012	0.0034 ± 0.0044	0.0348 ± 0.0015	-0.1489 ± 0.4359	4.3498 ± 1.9415
	4	-0.0011 ± 0.0023	0.0117 ± 0.0057	0.0473 ± 0.0025	-0.9589 ± 0.3214	3.5843 ± 1.3078
	5	0.0024 ± 0.0034	0.0177 ± 0.0060	0.0480 ± 0.0024	-0.4726 ± 0.1663	-0.0473 ± 0.3678
	6	0.0054 ± 0.0065	0.0243 ± 0.0051	0.0600 ± 0.0044	-0.6214 ± 0.2125	-0.0186 ± 0.4853
	7	0.0075 ± 0.0094	0.0300 ± 0.0081	0.0709 ± 0.0119	-1.6363 ± 0.7897	5.4609 ± 3.3148
	8	0.0245 ± 0.0106	0.0431 ± 0.0071	0.0617 ± 0.0071	-1.0113 ± 0.3429	-0.2420 ± 1.0925
	9	0.0116 ± 0.0184	0.0415 ± 0.0327	0.0773 ± 0.0088	-0.4632 ± 0.4136	-1.1720 ± 0.7070
	10	0.0199 ± 0.0264	0.0406 ± 0.0181	0.0782 ± 0.0305	-1.8905 ± 0.7651	2.3650 ± 1.7286

Table B.7: Estadísticas multiescala de AllTReturns para GBM. Cada entrada es estimador ± error estándar (4 decimales).

Index	duration	mean ± err	median ± err	std dev ± err	skewness ± err	kurtosis ± err
	1	-0.0001 ± 0.0002	-0.0003 ± 0.0002	0.0095 ± 0.0001	0.0373 ± 0.0508	-0.0131 ± 0.0817
	2	0.0002 ± 0.0004	-0.0000 ± 0.0023	0.0171 ± 0.0002	0.0067 ± 0.0425	-0.9041 ± 0.0658
	3	0.0008 ± 0.0009	0.0039 ± 0.0058	0.0247 ± 0.0005	-0.0294 ± 0.0726	-1.2638 ± 0.0547
	4	0.0024 ± 0.0019	0.0106 ± 0.0096	0.0322 ± 0.0007	-0.1144 ± 0.0980	-1.4286 ± 0.0671
GBM	5	0.0025 ± 0.0025	0.0135 ± 0.0146	0.0398 ± 0.0011	-0.0947 ± 0.1234	-1.5313 ± 0.0817
	6	0.0047 ± 0.0051	0.0197 ± 0.0201	0.0473 ± 0.0016	-0.1571 ± 0.1820	-1.5917 ± 0.1236
	7	0.0085 ± 0.0086	0.0255 ± 0.0248	0.0542 ± 0.0027	-0.2856 ± 0.2900	-1.5319 ± 0.2965
	8	0.0050 ± 0.0133	0.0138 ± 0.0414	0.0620 ± 0.0038	-0.1241 ± 0.4488	-1.6456 ± 0.3702
	9	0.0172 ± 0.0044	0.0484 ± 0.0023	0.0702 ± 0.0010	-0.2675 ± 0.2140	-1.7509 ± 0.2586

Appendix C. GBM - Trends Duration Distribution

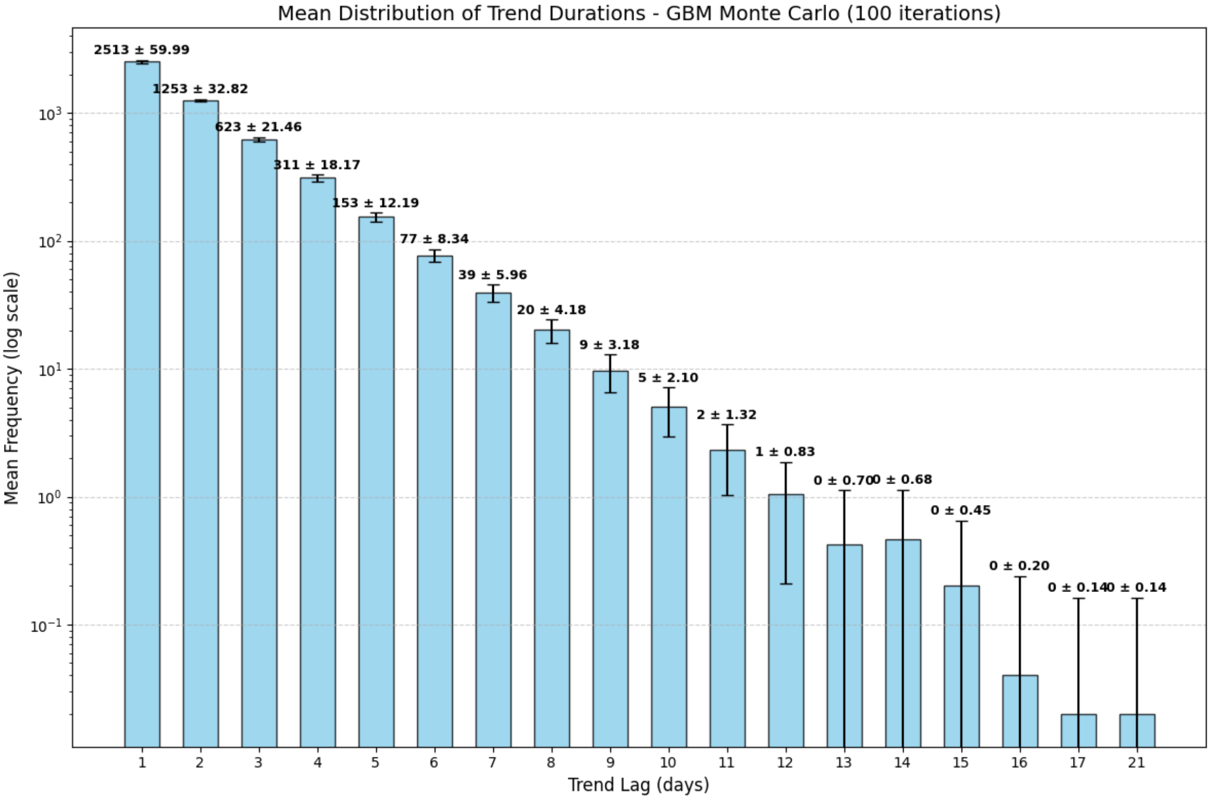

Figure C.11: Duration distribution GBM